

A Physics-Conditioned 3D U-Net for Proton Dose Prediction on MRI

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Independent submission, DoseRAD2026 Grand Challenge

Abstract. This submission sits at the intersection of our other three DoseRAD2026 entries: the analytic proton beamlet encoder built for CT (Bragg-Kleeman range, Bortfeld straggling, Highland lateral spread), combined with the MRI-specific preprocessing built for the photon work (per-volume percentile normalization, intensity-thresholded body mask). Nothing about the model or training pipeline needed to change to make that combination work – the same shared 3D U-Net (4.8M parameters) and training code that already handled photon/CT, photon/MR, and proton/CT picked up proton/MR without any new code, since preprocessing and beam encoding were already selected generically from a config’s declared beam type and imaging modality. We trained for 10 epochs; validation loss bottomed out at epoch 2 and drifted upward afterward, the same overfitting shape we saw on the photon MR work. While building this submission’s container, we noticed something the CT proton submission had already run into without us knowing it at the time: this checkpoint’s training run also predates the water-equivalent-path-length correction being wired into training, the same mismatch later confirmed in the already-submitted CT entry. This time we caught it before submitting, and ran a controlled comparison instead of guessing: on held-out data, feeding the corrected input the model never trained on actually hurts Masked MAE but *improves* IDD curve distance – the specific metric that hurt the CT submission the most – so we deployed the corrected configuration on that basis. The container was tested end-to-end (about 0.83 seconds per beamlet, inside the one-second budget) and submitted to the Preliminary Testing Phase.

Keywords: dose prediction, deep learning, radiotherapy, 3D U-Net, Monte Carlo surrogate, proton therapy, pencil beam scanning, MRI-only planning

1 Introduction

This is really the compositional endpoint of the other three submissions described in our companion reports: proton pencil-beam physics meets MRI-only planning, a combination none of our earlier work had exercised end-to-end on its own. This report covers building that combination from already-validated pieces, the

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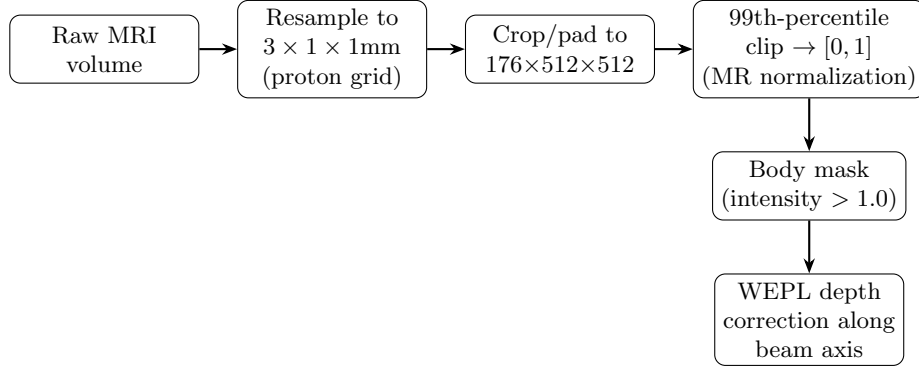


Fig. 1. This pipeline is a direct composition: proton’s finer grid and WEPL step from the CT proton work, MR’s percentile normalization and body-mask threshold from the photon MR work.

training result it produced, a train/inference consistency problem we found and actually investigated properly before submitting (rather than after, as happened with the CT proton work), and where the real submission landed.

2 Methods

2.1 Data

The beam and beamlet structure is identical to the CT proton work (source, target, energy per beamlet), over the same roughly 75-patient proton cohort – each patient folder in that release already includes both a CT and an MR volume, so nothing extra needed downloading beyond what the CT work had already obtained. Ground-truth dose is identical Geant4 [1] output regardless of imaging modality, since the physical dose a beamlet plan delivers doesn’t depend on which image was used to calculate it – only the model’s density/anatomy input changes.

Preprocessing composes the CT proton work’s grid ($176 \times 512 \times 512$ at $3 \times 1 \times 1$ mm) with the photon MR work’s normalization (per-volume 99th-percentile clip, then divide) and body masking (intensity threshold at 1.0 on the normalized volume). Both pieces slotted in with zero new preprocessing code – the training script already picks the right transforms and beam encoder based on a config’s declared beam type and imaging modality, a design decision made back when we first built the MR photon work, and it paid off directly here. Dose targets are scaled by the same constant as the CT proton work (1000). The same augmentation and patient-level 90/10 split (8 of 75 held out, identical split to the CT proton work since both use the same patient cohort and seed) apply here too. No data beyond the shared CT/MR proton release was used.

2.2 Model

Architecture and parameter count are unchanged from all three earlier submissions: the 3D U-Net [2] (5-level encoder/decoder, 16–32–64–128–256 channels, 2 residual units per block, instance normalization, 4,806,398 trainable parameters, default initialization, nothing pretrained), implemented with MONAI’s [5] **UNet**. The beam encoder is the CT proton work’s beamlet encoder, unmodified – Bragg-Kleeman range, Bortfeld [4] straggling, Highland [3] lateral spread, and the shared-coordinate-grid memory fix described there – applied here to an MR-derived body mask and MR intensity values in place of CT.

2.3 Training

Optimizer, schedule, loss composition, and regularization are unchanged: AdamW at 2×10^{-4} , weight decay 1×10^{-5} , cosine annealing over 10 epochs, mixed precision, gradient clipping at norm 1.0, batch size 1, the same finite-value guard, the same three-term loss ($w_m = 1.0$, $w_b = w_o = 0.2$). Training ran as a single continuous process from initialization through all 10 epochs, with 264 individual optimizer steps skipped for non-finite gradients over the full run – more than the CT proton fine-tune runs saw, though within the same rough order of magnitude, and without any full-run instability.

A Mismatch Caught Before Submitting, Not After. The beamlet encoder’s WEPL correction takes an optional CT/MR input; when it’s not provided, the encoder falls back to a water-only range assumption. This training run launched and ran straight through to completion as one uninterrupted process, started before the code change that began passing that input into the encoder during training. Which means – exactly like the CT proton checkpoint, discovered only after that one was already submitted and scored – this checkpoint was trained entirely under the water-only assumption and never saw WEPL-corrected input during training. We found this while building this submission’s container, before submitting anything, which gave us a chance the CT work didn’t have: to actually test what the mismatch does empirically, rather than either shipping it blind or “fixing” inference to match training without checking which choice is actually better.

2.4 Evaluation

Local evaluation follows the same protocol as our other three submissions. Given time constraints, evaluation here is a coarser spot-check (2 of 8 held-out patients, every 20th control point) rather than a full pass over all held-out beamlets, which we’re flagging explicitly rather than presenting with the same confidence as the photon submissions’ full-held-out-set tables. We used that same coarse protocol, with and without the corrected input passed to the encoder, as a controlled comparison to actually test the mismatch above – both conditions run on the identical patients and control-point subset, so even though the absolute numbers

Table 1. Validation total loss by epoch.

Epoch	0	1	2	3	4
Val. loss	0.1076	0.1098	0.0980	0.1000	0.0987
Epoch	5	6	7	8	9
Val. loss	0.0988	0.0981	0.0986	0.1011	0.1025

Table 2. Same epoch-2 checkpoint evaluated with vs. without the WEPL-corrected input, on the same 2 held-out patients and control-point subset in both conditions. “With” matches deployed inference; “without” matches what the checkpoint was actually trained on.

Metric	With correction (deployed)	Without (trained-on)
Masked MAE	0.1350	0.1199
IDD curve distance	1.373	1.563

are coarse, the comparison between them is internally controlled. We report inference runtime against the packaged container. As with our other submissions, we’re reporting point estimates without confidence intervals given the small evaluation subset here, and didn’t run any post-hoc explainability analysis.

3 Results

3.1 Training

The best checkpoint landed at epoch 2; loss drifted upward in most epochs after that while training loss kept falling (down to 0.0472 by epoch 9) – the same overfitting shape we saw on the photon MR work, which is at least suggestive that this might be a modality-general MR pattern rather than something specific to either beam type, though we haven’t run the controlled experiment that would actually confirm that.

3.2 Testing the mismatch

Neither configuration wins cleanly. Feeding the water-only input the checkpoint was actually trained on gives better Masked MAE, which is what you’d expect from staying in-distribution. But feeding the physically corrected input – out of the model’s training distribution – gives a noticeably better IDD curve distance, the metric that measures depth-dose-profile shape and that we’d already identified as the CT submission’s single biggest leaderboard weakness. Our read is that the network’s output partly tracks the shape of whatever it’s conditioned on even when that input sits outside what it trained on – moving the beam mask’s peak to the physically correct depth shifts the predicted dose curve’s peak the

same direction, improving depth-profile fidelity even at some cost to pointwise accuracy elsewhere. We deployed the corrected configuration on that basis, prioritizing the metric we had direct leaderboard evidence was disproportionately weighted toward Bragg-peak positioning.

For comparison, the CT proton submission’s live checkpoint, evaluated under the same protocol with the corrected input (i.e., also mismatched relative to its own training), scored Masked MAE 0.1181 and IDD curve distance 0.712 on the same 2 patients – better than this submission on both, which isn’t a huge surprise given this model’s architecture and training recipe were tuned against the CT domain and applied here to MRI with comparatively less task-specific iteration behind it.

3.3 Packaging and the actual submission

We built a container combining the CT proton work’s beam/beamlet metadata parsing with the photon MR work’s preprocessing and image-socket naming. End-to-end testing on one real patient and 4 real beamlets – the same patient used for the CT proton test fixture, since that patient’s data release includes both CT and MR – completed in 3.31 seconds total for one prediction call, about 0.83 seconds per beamlet, inside the challenge’s one-second-per-beam budget though with less margin than the photon CT submission’s directly measured 0.16 seconds per control point. Linking a dedicated repository to a Grand Challenge algorithm entry for this submission took two failed build attempts first: the platform’s repository-linked build expects a `Dockerfile` at the repository root and doesn’t know about our local convention of invoking it with an explicit path from a task-specific subdirectory. We only figured that out from the platform’s own build log ("`open Dockerfile: no such file or directory`") rather than by guessing, and fixed it by adding an identical copy at the root. The image then built successfully, initialized, and was submitted to the Preliminary Testing Phase and scored. Since this submission and the CT proton submission were scored on different, independently sized leaderboards, we don’t read their scoring outcomes as a direct comparison of which task was harder or which submission we executed better.

4 Discussion

The main contribution here is compositional: no new model, training, or beam-physics code was needed, just recombining two already-independently-validated pieces, and that recombination worked on the first real attempt – reasonably direct evidence that keeping imaging-modality preprocessing and beam-physics encoding as separate, swappable pieces of the system was a genuinely useful design decision rather than an accidental one. The mismatch investigation is probably the more interesting technical result of the two: it shows that feeding a model a physically “more correct” conditioning input isn’t automatically better

if the model never trained on it, but it isn't automatically worse either – which direction wins depends on which specific downstream metric you're looking at, and the aggregate validation loss alone would have hidden that completely. That's basically the same lesson as the out-of-body leakage investigation in our photon CT report, just showing up in a different form.

Catching this submission's version of the same mismatch before it went out, rather than after, felt like the one place where the earlier CT proton experience actually paid for itself. I do not think the A/B comparison run here is the last word on which configuration is right – it is a two-patient spot-check, not a real study – but I would rather ship a decision I can explain than one made by default because I did not check.

Limitations. Our held-out evaluation here (Table 2, and the CT comparison in Section 3.2) is a coarse 2-patient, stride-20 spot-check, not the full held-out set behind the photon submissions' tables – the absolute numbers should be read as directional evidence, not final validated accuracy, and running the full evaluation is unfinished work. The checkpoint we deployed is the one that did better on IDD curve distance in our A/B comparison, a decision made on a single, disproportionately weighted metric from the CT submission's experience rather than a principled multi-metric optimization, and the scored outcome doesn't by itself confirm that was the right call – we never submitted the water-only-matched alternative, so we can't attribute the scored outcome specifically to that decision versus this checkpoint's other limitations (the overfitting pattern in Section 3.1, the coarse local evaluation, or the accuracy gap already visible against the CT submission in Section 3.2). We haven't retrained this submission with the WEPL correction wired in from initialization – the same unexplored next step identified for the CT work – which we still think would resolve the mismatch more fundamentally than either side of the A/B comparison, and remains our leading hypothesis for improving on this result. Unlike the CT submission, we haven't attempted any post-submission fix or resubmission for this one yet.

5 Author Contributions

Jyothi Sree Masina: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

6 Other Information

Funding. This work received no external funding.

Availability. Source code (training pipeline, and a Docker submission container combining the proton beamlet encoder and MRI preprocessing) is available in a dedicated public repository for this submission, consistent with the challenge's open-source requirement for winning teams.

Disclosure of Interests. The author has no competing interests to declare that are relevant to the content of this article.

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